C# 1D Array Operations

Complete code collection • Beginner-friendly examples

Note: Each example is a separate complete C# program. For insertion examples, the array has one extra slot. Deletion examples use `size` to track active elements.

1. Array Traversal

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = { 10, 20, 30, 40, 50 };

        for (int i = 0; i < numbers.Length; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

2. Array Insertion at the End

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int newValue = 50;

        numbers[4] = newValue;

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

3. Array Insertion at the Beginning

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int newValue = 5;

        for (int i = 3; i >= 0; i--)
        {
            numbers[i + 1] = numbers[i];
        }

        numbers[0] = newValue;

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

4. Array Insertion at a Specific Position

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int position = 3;
        int newValue = 25;

        for (int i = 3; i >= position - 1; i--)
        {
            numbers[i + 1] = numbers[i];
        }

        numbers[position - 1] = newValue;

        for (int i = 0; i < 5; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

5. Array Deletion at the End

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int size = 4;

        // Delete at the end
        size--;

        // Display the array
        for (int i = 0; i < size; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

6. Array Deletion at the Beginning

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int size = 4;

        // Delete at the beginning
        for (int i = 0; i < size - 1; i++)
        {
            numbers[i] = numbers[i + 1];
        }

        size--;

        // Display the array
        for (int i = 0; i < size; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

7. Array Deletion at a Specific Position

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[5];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int size = 4;
        int position = 2;

        // Delete at a specific position
        for (int i = position - 1; i < size - 1; i++)
        {
            numbers[i] = numbers[i + 1];
        }

        size--;

        // Display the array
        for (int i = 0; i < size; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

8. Array Updation

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = new int[4];

        numbers[0] = 10;
        numbers[1] = 20;
        numbers[2] = 30;
        numbers[3] = 40;

        int position = 2;
        int newValue = 25;

        // Update the array
        numbers[position - 1] = newValue;

        // Display the array
        for (int i = 0; i < numbers.Length; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```

9. Array Searching (Linear Search)

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = { 10, 20, 30, 40, 50 };

        int searchValue = 30;
        bool found = false;

        for (int i = 0; i < numbers.Length; i++)
        {
            if (numbers[i] == searchValue)
            {
                Console.WriteLine("Value found at index " + i);
                found = true;
                break;
            }
        }

        if (found == false)
        {
            Console.WriteLine("Value not found");
        }
    }
}
```

10. Array Reset

```
using System;

class Program
{
    static void Main()
    {
        int[] numbers = { 10, 20, 30, 40 };

        // Reset the array
        for (int i = 0; i < numbers.Length; i++)
        {
            numbers[i] = 0;
        }

        // Display the array
        for (int i = 0; i < numbers.Length; i++)
        {
            Console.WriteLine(numbers[i]);
        }
    }
}
```